

For this LDO design project, we started by trying to understand what the most important aspects of the LDO block diagram are. For starters, we really wanted to have good Error amplification. To achieve this, we decided to build off of our Folded Cascode from MP4 since it was able to deliver very good gain and start off well above the phase margin. This design choice inherently leads to higher I_Q as the current used to bias the Folded Cascode is much higher than if we were to use smaller amplifier topologies.

Once our Error Amplifier was set up, we began to build the LDO circuit to match the schematic symbol provided. This meant adding the MPASS MOSFET, and the feedback resistors to the LDO cellview. When choosing the MPASS MOSFET, we chose to use a PMOS that was sized very large. The large size of the MPASS PMOS is due to its ability to drive a lot of current if needed, as well as maintain a low R_{on} . Keeping R_{on} low also allows for better load regulation as the output resistance is the dominant term to tune when trying to get the load regulation within a certain amount. On the topic of output resistance, the choice of feedback resistors was also crucial to getting the load regulation in check, as well as a better efficiency from the LDO. The feedback resistors we chose were two $3k\Omega$ resistor. This made it so our feedback branch did not draw a lot of current, while still allowing for a low output resistance, improving most of our LDO metrics. The sizing of feedback resistance made it so that the output pole was pushed much further out than the dominant pole, improving stability and improving I_Q .

Up to this point, the circuit was set up to for the LDO schematic provided, but an important problem still remained. The dominant pole was at the output of the LDO. This means that the stability of the output relies heavily on the output resistance, which furthermore means stability can change with varying loads at the output. To fix this, we introduced another capacitor that connects between the gate and drain of the MPASS PMOS. We chose it to have a lower capacitance, effectively creating a pole that is closer to the origin and making stability dependent on the output resistance of the error amplifier and not the load.

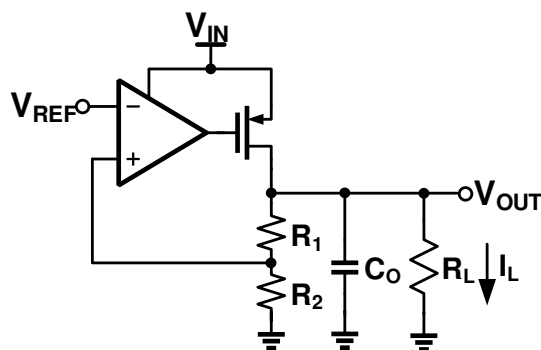


Figure 1: LDO schematic.

SCHEMATIC AND DEVICE PARAMETERS

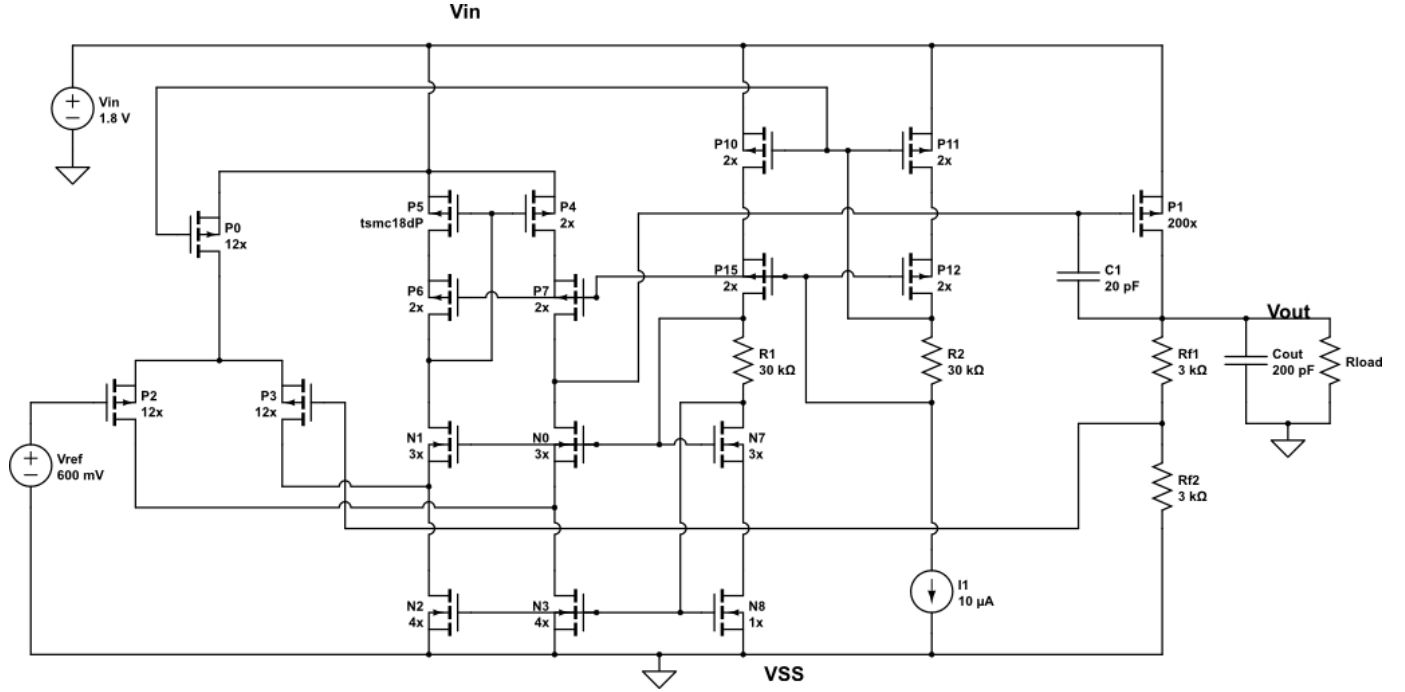


Figure 2: Folded Cascode with LDO.

Table 1: Device Parameters

Device	W (μ m)	L (nm)	M	g_m (S)	r_o (Ω)	V_{OV} (mV)	I_D (A)
P0	2.88	270	12	972.4 μ	86.12k	134	65.17 μ
P2	2.88	270	12	583.2 μ	148.7k	111.7	32.57 μ
P3	2.88	270	12	583.7 μ	148.5k	111.7	32.6 μ
P4	2.88	270	2	114.9 μ	455.5k	118.5	6.8 μ
P5	2.88	270	2	114.7 μ	448k	118.5	6.79 μ
P6	2.88	270	2	115.3 μ	582k	117.9	6.79 μ
P7	2.88	270	2	115.7 μ	713.3k	117.6	6.81 μ
P10	2.88	270	2	148.5 μ	270k	131.9	9.79 μ
P11	2.88	270	2	151.8 μ	375k	131.7	10 μ
P12	2.88	270	2	151.2 μ	394.5k	132.2	10 μ
P15	2.88	270	2	150.9 μ	569.8k	129.7	9.79 μ
P1	1.44	270	200	48.42m	971	421.2	10.2m
N0	1.44	580	3	125.9 μ	818.5k	108	6.8 μ
N1	1.44	580	3	125.7 μ	845.1k	108.1	6.79 μ
N2	1.44	580	4	440.6 μ	124.7k	178.8	39.4 μ
N3	1.44	580	4	440.3 μ	123.1k	178.8	39.37 μ
N7	1.44	580	3	165.3 μ	507.9k	118.4	9.79 μ
N8	1.44	580	1	109.4 μ	432.4k	179	9.79 μ

PERFORMANCE SUMMARY

Table 2: Performance summary

Design parameter/variable	Simulated performance	Specification
Input voltage	1.8V	$\leq 1.8\text{V}$
Output voltage	1.2V	1.0V – 1.4V
Load current	1mA	0.1mA – 10mA
DC load regulation	$13.8 \frac{\mu\text{V}}{\text{mA}}$	$\leq 50\mu\text{V}/\text{mA}$
DC line regulation	$55 \frac{\mu\text{V}}{\text{V}}$	$\leq 500\mu\text{V}/\text{V}$
Quiescent current ($I_L = 0.1\text{mA}/I_L = 10\text{mA}$)	298.6uA / 300uA	Minimum
PSR (@ $F_{in} = 1\text{KHz}/F_{in} = 1\text{MHz}$ @ 0.1mA)	-66dB / -6.2dB	–
PSR (@ $F_{in} = 1\text{KHz}/F_{in} = 1\text{MHz}$ @ 10mA)	-65dB / -7.1dB	–
Worst-case PSR	-6.12dB @ 1 MHz w/ 0.1mA	–
DC loop gain ($I_L = 0.1\text{mA}/I_L = 10\text{mA}$)	89.7dB / 78.75dB	–
Loop-gain unity gain frequency ($I_L = 0.1\text{mA}/I_L = 10\text{mA}$)	1.68MHz / 1.65MHz	–
Loop-gain phase margin($I_L = 0.1\text{mA}/I_L = 10\text{mA}$)	66.2° / 86.9°	–
Loop-gain gain margin ($I_L = 0.1\text{mA}/I_L = 10\text{mA}$)	25.28 / 36.77	–
Output noise ($I_L = 0.1\text{mA}/I_L = 10\text{mA}$)	$3.44 \frac{\text{fV}^2}{\text{Hz}}$ / $4.5 \frac{\text{fV}^2}{\text{Hz}}$	–

LOOP-GAIN AC RESPONSE

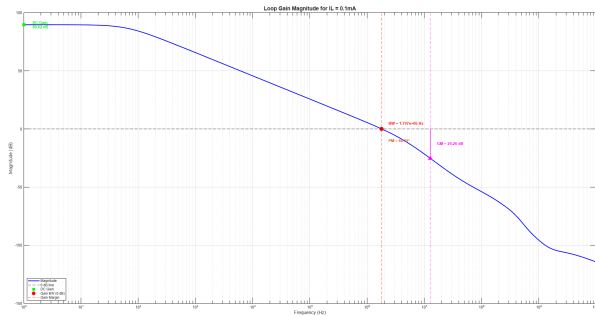


Figure 3: Loop Gain Magnitude: 0.1 mA.

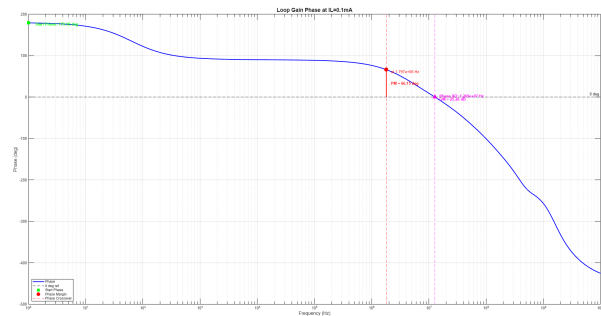


Figure 4: Loop Gain Phase: 0.1 mA.

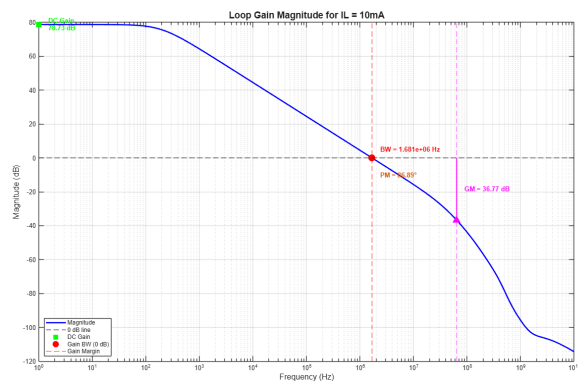


Figure 5: Loop Gain Magnitude: 10 mA.

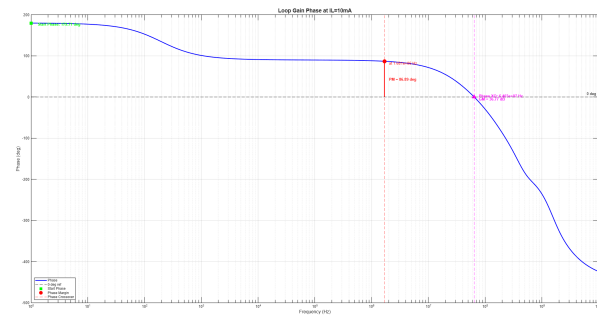


Figure 6: Loop Gain Phase: 10 mA.

DC LOAD AND LINE REGULATION RESPONSE

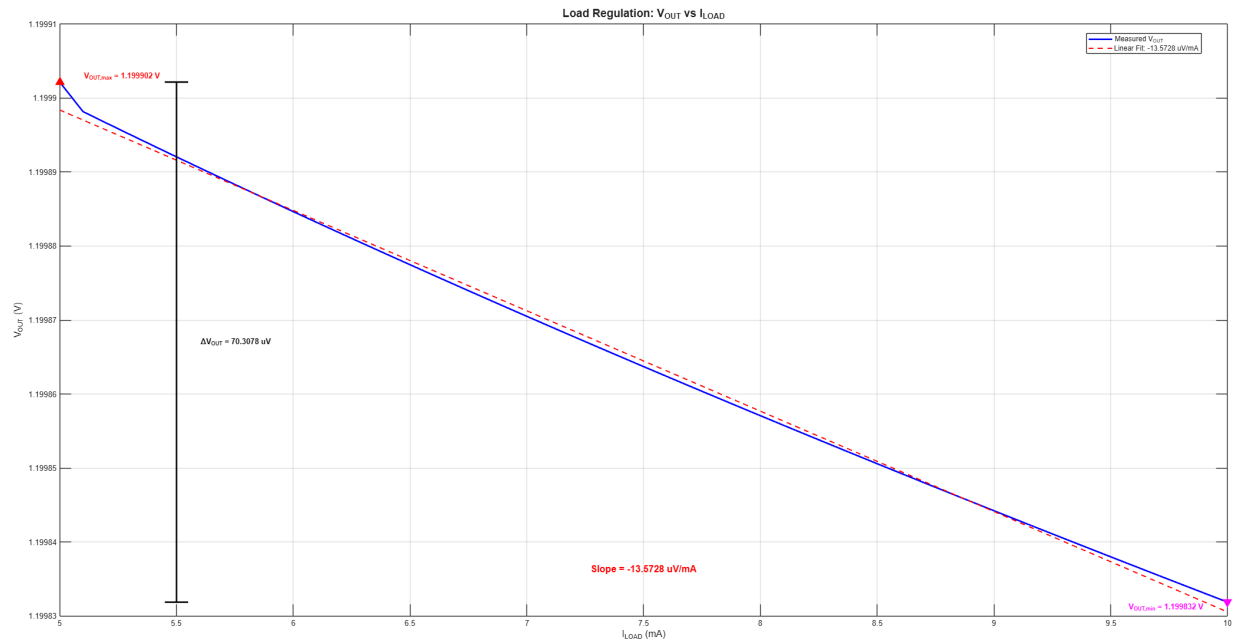


Figure 7: DC Load Regulation.

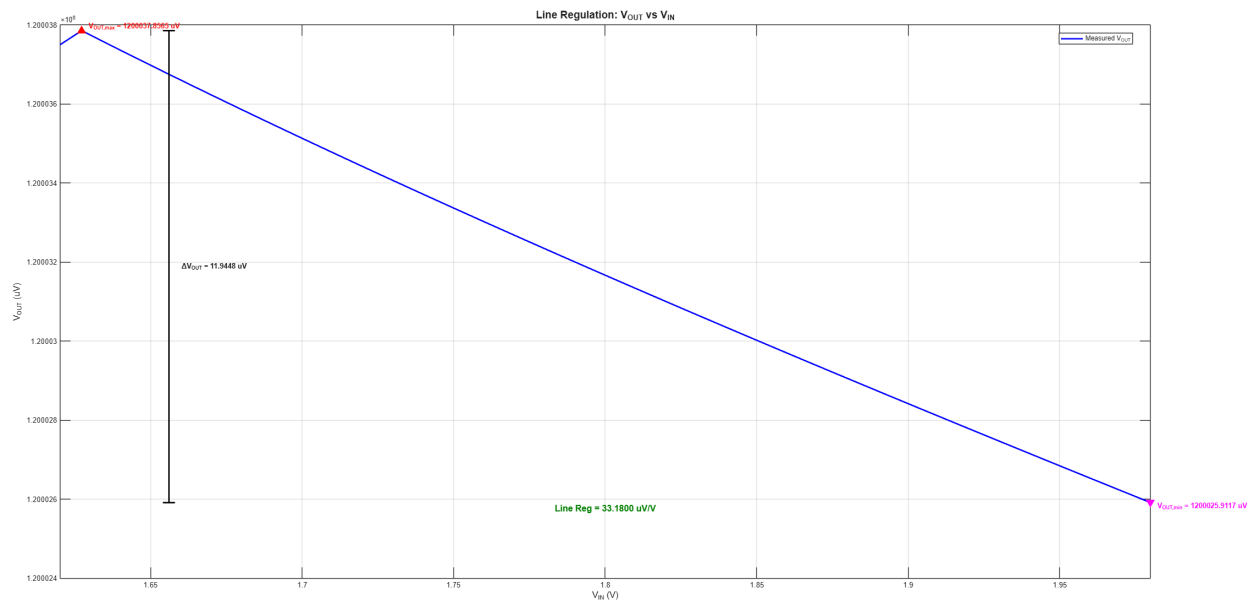


Figure 8: DC Line Regulation.

POWER SUPPLY REJECTION

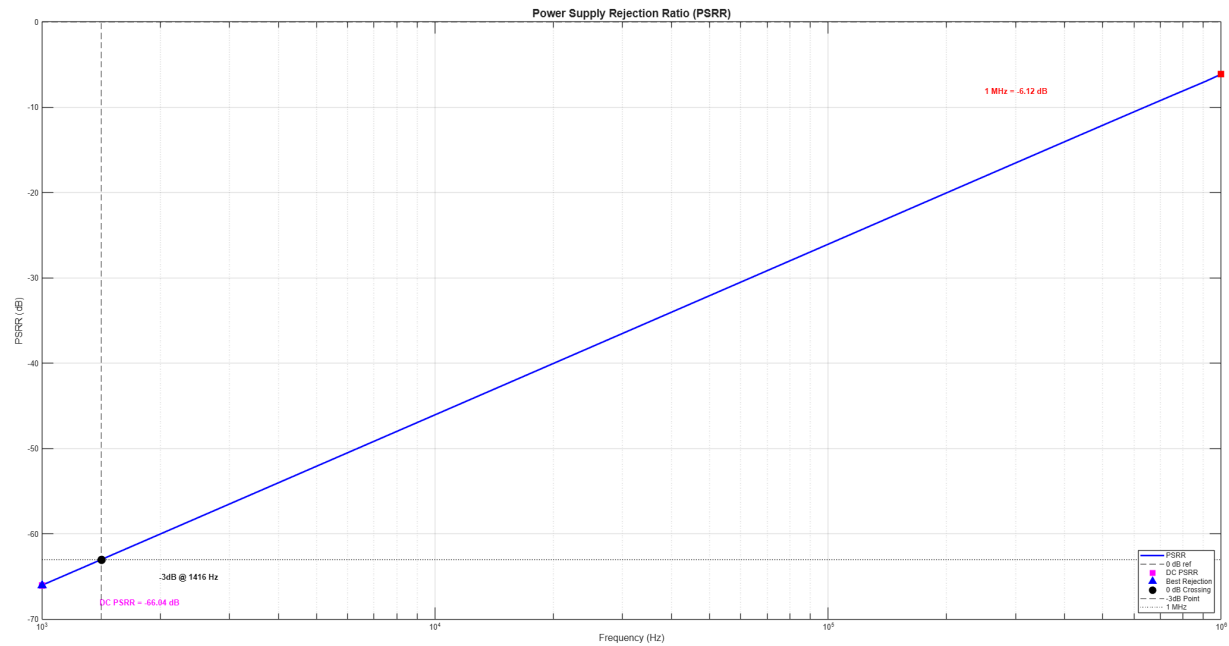


Figure 9: Power Supply Rejection: 0.1 mA.

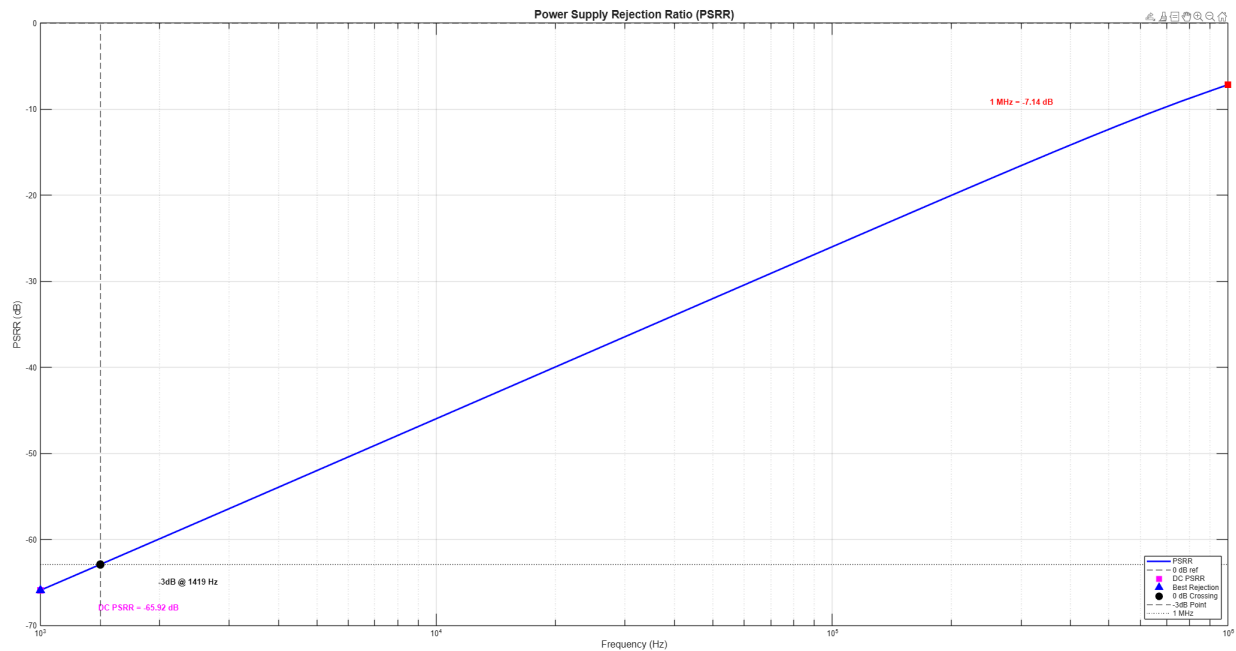


Figure 10: Power Supply Rejection: 10 mA.